

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Interview Trainer Agent

Domain of Project – Career Development / AI Interview Preparation

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Problem Statement

Problem Statement: Interview Trainer Agent

The Challenge

Job interviews are a critical step in the hiring process, yet many candidates struggle with effective preparation, confidence building, and answering questions in a structured manner. Interview resources, role-specific questions, company expectations, and preparation strategies are often scattered across multiple platforms, making it difficult for candidates to prepare efficiently.

Without personalized guidance and realistic practice, candidates may find it challenging to identify their weaknesses, improve communication skills, and perform confidently during interviews.

The Objective

Build an **AI-powered Interview Trainer Agent** that delivers personalized interview preparation and guidance for job seekers across different industries and experience levels. The solution should:

- **Personalized Interview Preparation-** Generate tailored interview questions based on the user's resume, experience level, skills, and target job role.
- **Interview Knowledge RAG Retrieval-** Retrieve and summarize role-specific interview questions, industry expectations, HR guidelines, and company interview patterns from trusted recruitment and professional sources.
- **Answer Evaluation & Feedback-** Analyze candidate responses and provide real-time feedback, scoring, improvement suggestions, and model answers.
- **Technical & Soft Skill Assessment-** Support both technical interviews and behavioral/HR interview preparation to ensure comprehensive readiness.

Multi-Agent System

- **Question Generation Agent** - Generates role-specific interview questions
- **Interview Evaluation Agent** - Analyzes answers and provides scoring
- **Feedback & Recommendation Agent** - Offers improvement tips and model responses
- **Knowledge Retrieval Agent** - Fetches interview-related information using RAG
- **Interview Analytics Dashboard** - Display interview scores, strengths, weaknesses, performance trends, and personalized preparation recommendations to help users improve over time.

Technology Used

Tech Stack Used

- Node.js
- Express.js
- HTML
- CSS
- JavaScript
- IBM Watsonx.ai
- IBM Granite Models
- Retrieval-Augmented Generation (RAG)
- PDF Resume Parsing
- JSON Knowledge Base

Approach Used

LangFlow + IBM Watsonx.ai + IBM Granite Models + RAG Architecture

Proposed Solution -

Proposed Solution: AI-Powered Interview Trainer Agent

The proposed solution is a multi-agent **AI-powered Interview Trainer Agent** built using **LangFlow**, **IBM watsonx.ai**, **IBM Granite Models**, and **RAG** to provide personalized interview preparation and guidance. The system helps candidates improve interview skills, build confidence, and receive real-time feedback.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to retrieve role-specific interview questions, industry expectations, HR guidelines, and technical concepts from trusted sources. This enables the Granite model to generate accurate questions, model answers, and preparation strategies.

The system consists of four key agents:

- **Interview Knowledge Agent** – Retrieves interview-related information using RAG.
- **Question Generation Agent** – Generates personalized questions based on the user's resume, experience, and job role.
- **Answer Evaluation Agent** – Evaluates candidate responses and provides scores.
- **Feedback & Recommendation Agent** – Suggests improvements, model answers, and preparation tips.

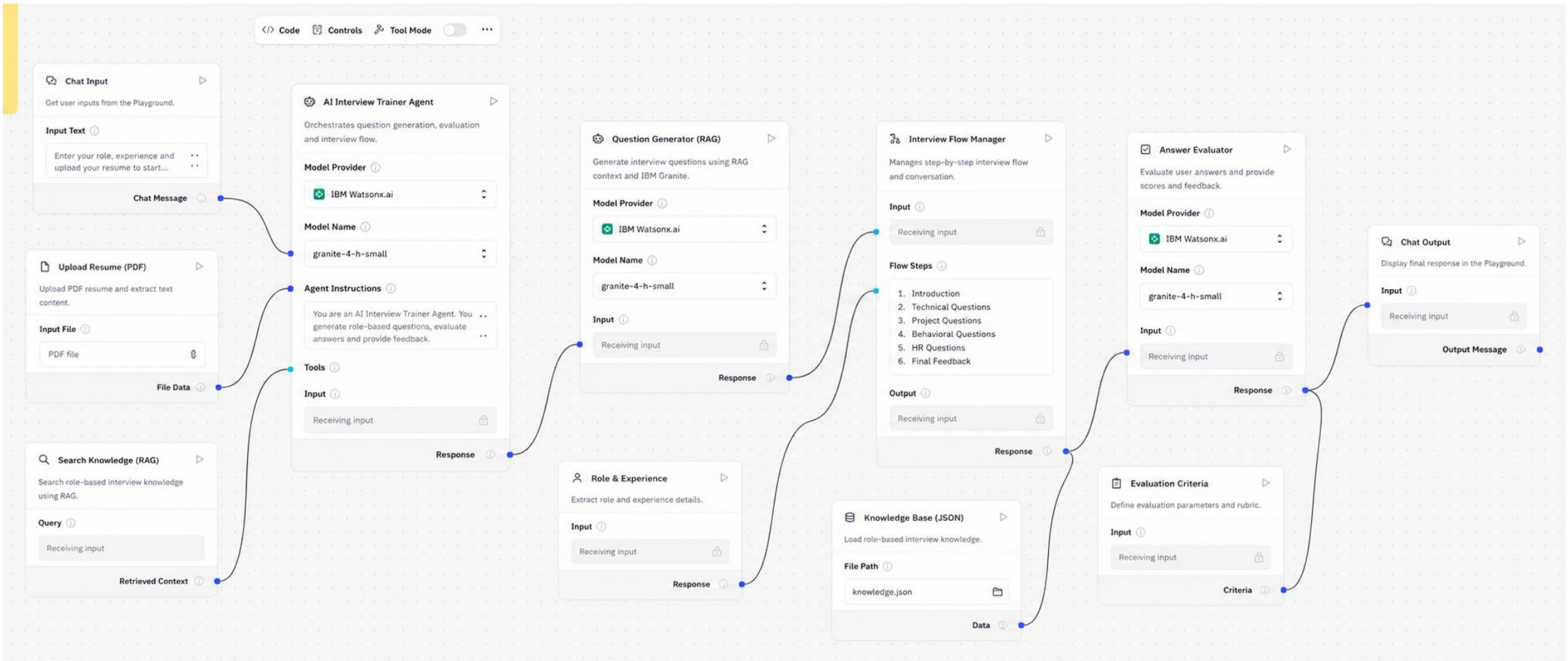
The solution includes a user-friendly dashboard that displays interview scores, strengths, weaknesses, and performance insights. **LangFlow** orchestrates the agent workflow, **watsonx.ai** manages model deployment and governance, and **IBM Granite Models** power question generation, evaluation, and feedback.

Overall, the solution delivers an intelligent interview preparation platform that simulates real interview experiences and helps candidates improve their chances of success in competitive hiring environments.

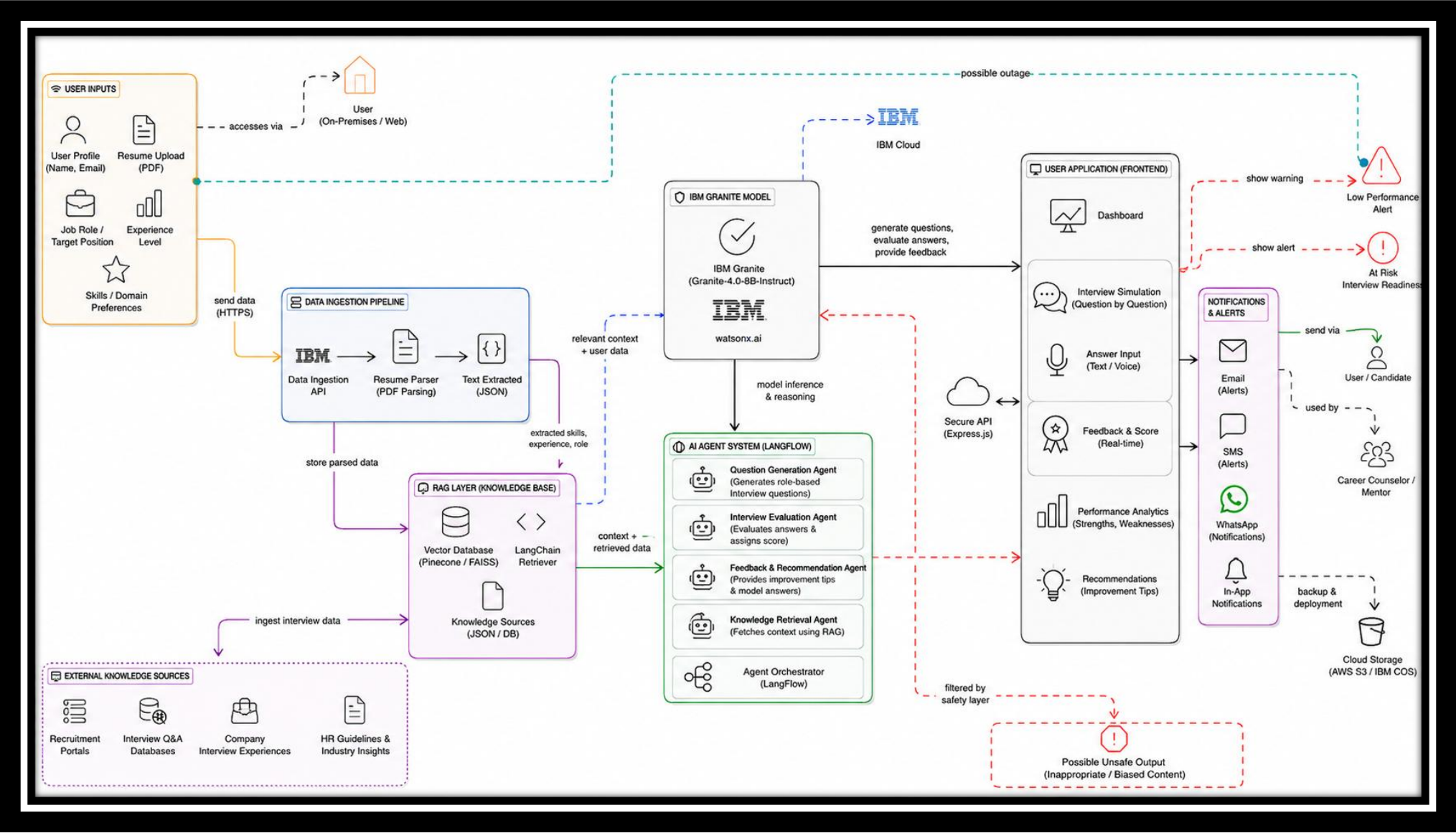
Langflow component Used -

- 1) **Chat Input** - Collects candidate details like job role, experience, and interview responses.
Used to start and personalize the interview session.
- 2) **IBM watsonx AI Agent** - Processes user inputs and manages question generation and evaluation flow.
Acts as the central controller for the interview system.
- 3) **IBM Granite Model** - Generates interview questions and evaluates candidate answers.
Provides recommendations based on performance.
- 4) **Chat Output** - Displays questions, scores, feedback, and preparation tips.
Presents the final response to the user in chat format.
- 5) **PDF Resume Upload** - Extracts key information from uploaded resumes.
Helps personalize interview questions and feedback.
- 6) **RAG Retriever** - Fetches relevant interview content from the knowledge base.
Ensures role-specific and accurate question generation.
- 7) **Interview Flow Manager** - Controls the step-by-step flow of the interview process.
Manages transitions from start to final feedback.
- 8) **Answer Evaluation Module** - Analyzes user answers for correctness and quality.
Returns scores, strengths, weaknesses, and improvements.
- 9) **Knowledge Base (JSON)** - Stores technical and behavioral questions with industry data.
Used by RAG to generate relevant interview content.

Langflow workflow -



Architecture blueprint



Role of Agentic AI in the solution

Role of Agentic AI in Interview Trainer Agent

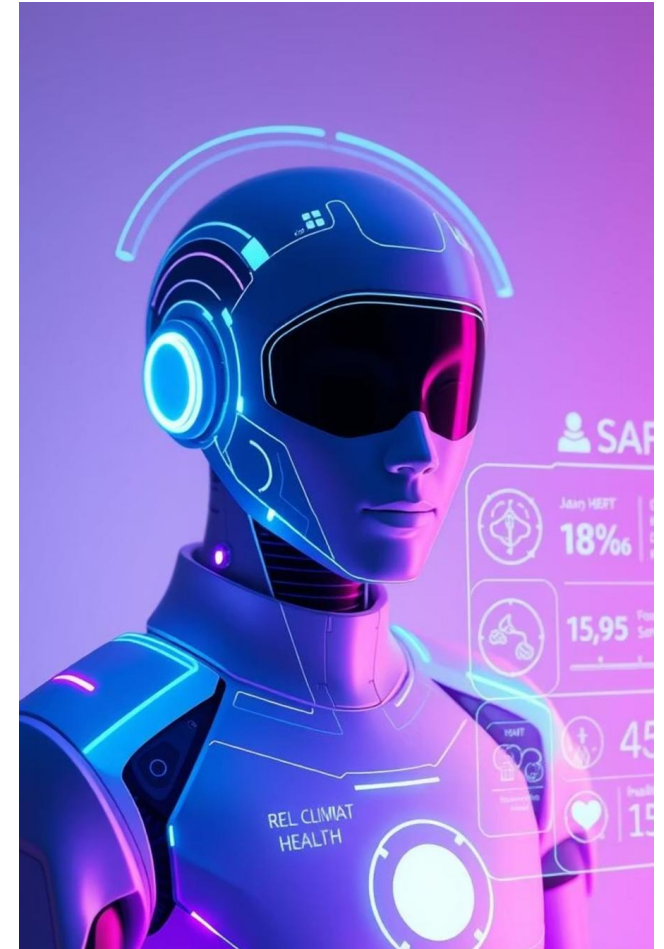
Agentic AI enables the **Interview Trainer Agent** to function as an intelligent, autonomous interview coach rather than a simple question-answer chatbot. Using **IBM watsonx.ai** and **IBM Granite Models**, it coordinates multiple specialized agents to deliver personalized interview preparation and guidance.

Each agent performs a specific role—such as retrieving interview knowledge, generating role-specific questions, evaluating candidate answers, and providing feedback—while working collaboratively. This multi-agent approach ensures accurate, relevant, and adaptive interview training.

Agentic AI also brings **context awareness**, understanding user details such as resume content, experience level, skills, and target job role to generate personalized interview questions and recommendations. It supports **real-time adaptation**, adjusting question difficulty and feedback based on candidate performance.

Additionally, the agents communicate with one another, enabling better decision-making, comprehensive evaluation, and continuous improvement throughout the interview session.

Overall, Agentic AI makes the system **proactive, personalized, and goal-driven**, acting like a smart virtual interviewer that helps candidates build confidence, improve responses, and increase their chances of success in competitive hiring environments.



Technology Used

- **LangFlow Platform** – For designing and orchestrating multi-agent interview workflows visually.
- **IBM Granite Model (Granite-4.0-8B-Instruct)** – For interview question generation, answer evaluation, reasoning, and personalized feedback.
- **IBM Cloud** – Provides the infrastructure to build, deploy, and scale the Interview Trainer Agent in a secure and reliable environment.
- **RAG (Retrieval-Augmented Generation)** – Retrieves role-specific interview questions, HR guidelines, industry expectations, and company interview insights before generating intelligent responses.
- **IBM Watsonx.ai** – For model deployment, prompt orchestration, embeddings, and AI governance.
- **Vector Database (FAISS/Chroma)** – Stores and retrieves interview knowledge, company interview experiences, and recruitment-related information for efficient RAG-based retrieval.
- **Node.js & Express.js** – Used for backend API development, workflow management, and communication between frontend and AI services.
- **PDF Resume Parser** – Extracts candidate skills, experience, education, and project details from uploaded resumes for personalized interview preparation.

Novelty and Uniqueness

The proposed **Interview Trainer Agent** stands out by combining **Agentic AI, RAG, and personalized interview coaching** into a single intelligent platform.

1. **Multi-Agent Intelligence** – Instead of a simple interview chatbot, multiple specialized agents collaborate to generate questions, evaluate answers, retrieve interview knowledge, and provide personalized feedback.
2. **RAG-Based Accuracy** – Integrates real-time interview knowledge retrieval with AI-generated responses, ensuring relevant, accurate, and role-specific interview preparation.
3. **Resume-Based Personalization** – Tailors interview questions, preparation strategies, and feedback based on the candidate's resume, skills, experience level, and target job role.
4. **Adaptive Interview Flow** – Dynamically adjusts question difficulty and interview progression based on candidate performance and responses.
5. **Real-Time Feedback & Scoring** – Instantly evaluates answers and provides scores, improvement suggestions, and model responses to accelerate learning.
6. **Comprehensive Assessment** – Supports both technical and behavioral/HR interview preparation, providing a complete interview readiness experience.
7. **Scalable & Enterprise-Ready** – Built on **IBM watsonx.ai** and **IBM Granite Models**, ensuring reliability, scalability, security, and AI governance.

This makes the solution a **smart, adaptive, and proactive virtual interview coach**, not just a static interview question generator.

Git Hub Link

Public GitHub Repository

Repository Name: AI-Interview-Trainer-Agent

Uploaded Files:

app.json

ProblemStatement22.pdf

ProjectPresentation.pptx

Frontend Source Code

Backend Source Code

README.md

GitHub Repository URL:

<https://github.com/AnjuK2006/Interview-Trainer-Agent->

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Future Scope

1. AI-Powered Voice Interview System –

The system can be enhanced with speech recognition and voice interaction capabilities, allowing users to participate in mock interviews through natural conversations. This will create a more realistic interview environment and help candidates improve communication, pronunciation, and confidence.

2. AI Interviewer Avatar & Multilingual Support –

Integrating a virtual AI interviewer avatar with support for multiple languages will make the platform more interactive and accessible. Users can practice interviews in their preferred language, improving inclusivity and helping candidates prepare for both local and global job opportunities.

3. Webcam-Based Interview Analysis –

The system can analyze facial expressions, eye contact, and body language during mock interviews to provide insights on confidence, professionalism, and communication skills.

4. Advanced Performance Analytics Dashboard –

Future versions can include detailed analytics, skill-gap identification, progress tracking, and personalized learning recommendations to help users continuously improve their interview performance.

5. Company-Specific Interview Preparation –

The platform can be expanded to provide interview questions, preparation strategies, and insights tailored to specific companies and industries, increasing candidate readiness for targeted job applications.

Thank You!

Thank you for your time and interest.